THE SCIENCE BEHIND COXEVAC CALCULATOR

Contents

Introduction

Q fever is an infectious disease common to many animal species including mammals, reptiles and birds. It is caused by a small intracellular bacterium: *Coxiella burnetii*. This disease is also a zoonosis and domestic ruminants (goats, sheep and cattle) are considered the reservoir of the bacteria and the main source of contamination for humans.

But beyond the zoonotic aspect, **Q fever is a major disease of ruminant farming**. Indeed, it is the cause of major health problems, linked to reproduction. The clinical expression is dominated in small ruminants by abortions, while in cattle other more insidious clinical signs, such as metritis or endometritis, placental retention and more generally fertility disorders are also consequences of the infection. Consequently, the impact of the disease on the profitability of the farm is difficult to assess.

Solutions to control the disease exist, in particular vaccination. As with any disease, the implementation of a vaccination protocol involves a financial and human investment. The farmer and his veterinarian legitimately need a tool to evaluate the profitability of vaccination in the short and medium term in order to make an informed decision about the implementation of a medical prophylaxis.

We have therefore developed this application to:

- **Estimate the cost of the disease in a given dairy herd,**
- **Determine the economic benefit of vaccination over 3 years and the return on investment that it allows.**

To do this, we worked with two independent experts:

Professor Didier Raboisson
of the National Veterinary
School of Toulouse, France.

Professor in Population Medicine & Economics of Animal Health

Professor Raphaël Guatteo
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France.

EBVS® European Specialist in Bovine Health Management

1. Economic cost of the disease

Impact of the disease in infected animals

The economic impact of Q fever is mostly based on reproductive disorders: extra-abortions, increase in placental retentions and metritis and decrease in reproduction performances:

For abortion

according to the literature, extra risk (relative risk) in cases of Q fever infection ranges from **2 to 2.5** [1.7-3.8] *(Lopez Gatius et al., 2012; Ordronneau, 2012)*.

Within a herd, it concerns both cows and heifers.

For placental retention

extra risk (relative risk) has been assessed at **1.52** [1.06-2.19] *(Ordronneau, 2012)*.

It concerns only cows since it can happen only after a first calving.

For metritis

extra risk (relative risk) has been defined in a field study at **2.5** *(Valla et al., 2014a)*.

Like for placental retention, it concerns only cows.

For other reproductive disorders

the following parameters have been defined by extrapolation from the literature defining the benefit of Q fever vaccination *(Lopez Gatius et al., 2012; Lopez Helguera et al., 2014)*.

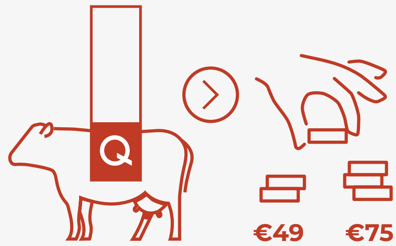
- For late service after first service, the extra risk (relative risk) in case of Q fever infection is **1.85** for both heifers and cows.
- According to literature, Q fever is responsible for a decrease by **50%** of the First Service Conception Rate (FSCR) for both cows and heifers.
- For extra service per conception: the same study evaluated it at **0.4**. It is applied to seropositive cows and heifers.
- Increase in number of days open (extra days in calving-to-calving interval (CCI) for cows or delayed first calving (C1) for heifers) directly linked to Q fever infection (i.e. independent of late service and for FSCR, as previously mentioned) is assessed at **14**.

Related costs

Data from the literature give the following costs for each parameter

- Mean cost of an abortion: **400** to **1000 euros** *(Arsia, 2010)*
- Cost of a day open: **5 euros** *(Cabrera, 2014)*
- Cost of an artificial insemination: highly variable depending on the country. For example: **30** to **40 euros in most of European countries**
- Cost of the treatment of a placental retention: **10** to **15 euros**
- Cost of the treatment of metritis / clinical endometritis: **40** to **100 euros**

Taking these data, for a herd with a seroprevalence of **30%** (a proportion frequently found in infected herds in Europe), the cost of the disease is between **€49** and **€75** per lactating cow per year.

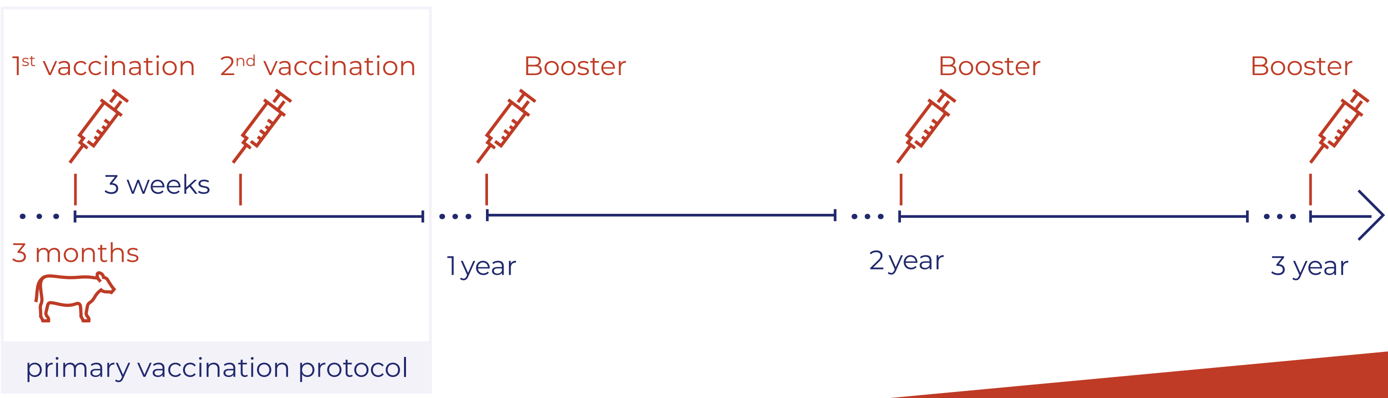


2. Vaccination with Coxevac

Chosen protocol

The chosen protocol for vaccination against Q fever is as follows:

- 1st year:
  - Vaccination of all animals present over 3 months of age
  - 2 injections 3 weeks apart (primary vaccination protocol)
- 2nd and 3rd year:
  - Annual booster for animals already vaccinated the year before
  - Primary vaccination protocol for young animals over 3 months of age and never vaccinated before

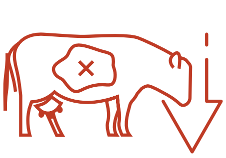


This protocol differs slightly from the one defined in the Coxevac specifications (booster 9 months after primary vaccination in cattle). However, this is the protocol widely used in the field. In addition, it has been shown that a booster 12 months after the two primary injections resulted in a similar increase in antibodies levels as a booster 9 months after primary vaccination *(Valla et al., 2014b)*.

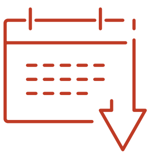
Benefit of vaccination

The vaccination against Q fever with Coxevac lower the risk for non-infected animals vaccinated when non-pregnant to become shedder (5 times lower probability in comparison with animals receiving a placebo), and reduce shedding of *Coxiella burnetii* in these animals via milk and vaginal mucus. This ultimately leads to the suppression of infection in the herd and thus to the elimination of all negative effects of Q fever.

In addition, for several years, numerous studies have been conducted on the health and zoo-technical benefits of Coxevac in cattle. In particular, it has been shown that vaccination allows:

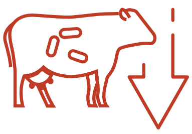


31% reduction in the risk of abortion due to Q fever
(Ordronneau, 2012)

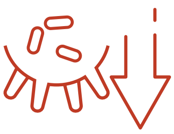


14-day reduction in calving-to-conception interval
(Lopez-Helguera et al., 2014)

39% increase (41.9/30.1 = 1.39) in conception rate at 1^{st} postpartum artificial insemination
(Lopez-Helguera et al., 2014)

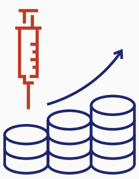


79% reduction in the risk of an animal becoming a shedder of the bacteria



No chronic shedding in milk for heifers
(Boettcher et al., 2017)

Elimination of environmental contamination (dust, aerosols and slurry) in less than two years
(Pinero et al., 2014)



From an economic point of view, Raboisson *et al.* (2022) showed that vaccination resulted in a **profit over 3 years** ranging from **3,000** to almost **12,000 euros** (depending on the initial infection rate) for a herd of 100 cows.

3. Calculator

Warnings

Only the parameters described in the previous two sections were taken into account for the development of the app.

We have deliberately chosen not to include the increase in culling rate due to the various reproductive disorders since, to our knowledge, there is no published data on this point specifically in relation to Q fever. Furthermore, although it has been described in the literature that Q fever can also be responsible for respiratory disease *(Plommet et al., 1973)*, mastitis or reduced milk production in cattle *(Barlow et al., 2008; Khatun et al., 2022)*, we did not include these data.

Overall, therefore, we assume that the tool developed underestimates the financial impact of the disease on a farm and, consequently, the benefit of implementing vaccination. We have chosen to focus the simulator's interest on the cattle farm where the disease was diagnosed. The overall economic impact of the disease, both in terms of the health of the surrounding farms and in terms of public health, was therefore not taken into account.

Finally, for simplicity, the model was built on the basis of a three-year period. However, vaccination should be extended beyond this period. Indeed, an early termination of the vaccination protocol leads to a re-emergence of the disease *(Courcoul et al., 2011)*.

Strategy of analysis

▸ Overview

The economic benefit of the vaccination is assessed through a partial budget analysis based on the difference between the situation with vaccination and without vaccination. Preliminary step is to evaluate the total cost of Q fever before vaccination i.e., production losses.

The production losses before (B) vaccination are equal to the total cost when no vaccination for one year, and is constant during a given analysis, years after years, assuming no change in the situation if no vaccination is done. The production losses after (A) vaccination are the final impact of Q fever, when vaccination is done (damage is partly decreased) for the 3 years after vaccination. It is considered that vaccination improved health and performances differently for cows and heifers (see calibration detailed on page 10).

The vaccine costs (V) are different for year 1 and for year 2 and 3 as primary vaccination is given to all the animals in the first year. In contrast, in year 2 and 3, animals already vaccinated in the previous year(s) simply receive a booster (as described on page 5).

The vaccination profit is defined as (B) - (A) – (V) for the 3-years period.

A partial budget approach has been judged appropriate, in spite this method only provides limited consideration of disease dynamics and of the interactions between components. The data available for calibration remains scarce and authors are convinced that a detailed dynamic and stochastic model won't bring any important added value due to the impossibility of adequate calibration of such complex model. The partial budget model is built at the herd level, independently for each year (i.e. static for a given year).

▸ Seroprevalence as a proxy of infection and its change if vaccination

The Q fever impact and the benefits of the vaccination are based on change in the seropositivity rate in the herd. In spite seropositivity is a bad indicator of infectious status of the cow, it is considered as a proxy of the herd level infection rate: seropositive cows are impacted by the Q fever and seronegative cows are not considered to be impacted by Q fever. In the simplified model, vaccination is considered to prevent seroconversions of seronegative cows (i.e. increases seronegative rate of the herd- a vaccinated cow remains seronegative i.e. diseases free) and the benefit increases year after year through reduced impact permitted by seroprevalence decreases.

In details, at the herd level, the seropositivity rate changes as follows:

In case of no vaccination and Q fever infection

- The prevalence is considered constant, culling of seropositive cows being compensated by new contamination of cows and heifers.

In case of vaccination in a herd with previously demonstrated Q fever circulation

- Seronegative vaccinated animals (cows or heifers, depending on the scenario) cannot become seropositive and the damage does not apply on them
- Culling of seropositive cows allows a reduction of the herd seroprevalence years after years
- Non vaccinated seronegative cows can become seropositive following the contamination rate (previously defined in case of no vaccination)
- These rules are applied independently for each year, based on seroprevalence observed end of previous year. An average yearly prevalence is defined based on the seroprevalence for Q fever at the beginning and the end of the year: the impact is applied to the average yearly seroprevalence. The method allows to consider herd dynamics and middle term change in prevalence in successive yearly partial budget assessments.

▸ Disease impact without vaccination

The disease impact is based on extra-abortions, increase in placental retentions and metritis and decrease in reproduction performances due to Q fever. The change in reproduction performances are based on:

- decrease in the first service conception rate (FSCR)
- extra service per conception
- extra days for conception (increase in calving to conception interval = extra days open).

The change in performances is different for heifers and cows when appropriate (see calibration detailed on page 10).

In details, the production losses are calculated as follows only for seropositive heifers and cows:

For abortion

extra risk (relative risk) in cases of Q fever infection is applied to base risk for abortion. It concerns both cows and heifers. According to the literature, **the Relative Risk (RR) ranges from 2 to 2.5** [1.7-3.8] *(Lopez Gatius et al., 2012; Ordronneau, 2012)*

For placental retention

extra risk (relative risk) is applied to base risk for placental retention and concerns only cows since it can happen only after a first calving. Retained placenta happening after an abortion in heifers are therefore not taken into consideration for this evaluation. **This RR has been assessed at 1.52** [1.06-2.19] *(Ordronneau, 2012)*

For metritis

extra risk (relative risk) is applied to base risk for metritis and like for placental retention, only for cows. **This RR has been defined in a field study at 2.5** *(Valla et al., 2014a)*

For other reproductive disorders

the following parameters have been defined by extrapolation from the literature defining the benefit of Q fever vaccination *(Lopez Gatius et al., 2012; Lopez Helguera et al., 2014)*.

- For late service after first service, the extra risk (relative risk) in case of Q fever infection is **1.85 for both heifers and cows**. It is transformed in 2 extra days for first calving 1 (C1) for heifers or 2 extra days in calving to calving interval for cows (CCI).
- For FSCR, the change in rate is also transformed into extra days of CCI ; According to literature, Q fever is responsible for a decrease by **50%** of FSCR value can be expressed as 3 days extra CCI per seropositive cow or 3 days at first calving per seropositive heifer.
- For extra service per conception: the same study evaluated it at **0.4**. It is applied to seropositive cows and heifers.
- Increase in number of days open (extra CCI or delayed C1) directly linked to Q fever infection (i.e. independent of late service and for FSCR, as previously mentioned) is assessed at **14**.

▸ Disease impact with vaccination

The vaccination is expected to take place in a herd where Q fever have been detected. Year 1 starts when vaccination is done and benefits are expected as soon as in year one.

Vaccine is seen as a control of damage that helps in the limitation of abortions and reproduction performances deterioration. Although it has been shown that Q fever can be responsible for placental retention and metritis/endometritis *(Valla et al., 2014a, Ordronneau, 2012, Dobos et al., 2020, De Biase et al., 2018)*, vaccine is not expected to reduce placental retention or metritis due to Q fever *(Ordronneau, 2012)*. **The benefits of vaccination rely:**

- **on the decrease in Q fever seroprevalence with time** for all the components, what is accounted for with the herd dynamics miming. For seropositive cows, the impact is defined similarly in situation with or without vaccination;
- **on extra benefit of vaccination on cows previously affected for abortion only**, with decreased risk (relative risk) of abortion if vaccinated compared to no vaccine for seropositive cows.

Calibration

The prevalence of positive cows in a dairy herd is known to be 40% under French conditions with ranges between 20-60% or 25-55% ; the national mean prevalence is 20% *(Taurel et al., 2011)*. However, if we consider the prevalence (assessed by PCR) in infected herds only, it varies from 19 to 39% *(Guatteo et al. 2011)*. As some seropositive animals do not shed *Coxiella burnetii*, prevalence based on PCR results is lower than seroprevalence. Consequently, in the absence of farm-specific data, for practical use, it is recommended to consider a seroprevalence of 30 % as default choice, and 40% in case of substantial number of abortions. Abortion rate can be considered as a good proxy that refer the herd Q fever infectious situation.

Impact of Q fever on heifers is applied on the same percentage (i.e. 30% or 40%) of animals, in spite evidence of lower seroprevalence in heifers compared to cows is available. To adjust for this, **the impact for heifer was considered as half the impact for cows, for both production losses and vaccination benefit**. This assumption allows to use one seroprevalence for the whole model calibration.

The non-modifiable fields are based on the bibliographic data as described above and reported in the table hereafter (Table 1).

	Q fever infection (EXTRA RISK COMPARED TO NO INFECTION)	Q fever vaccination (REDUCED RISK COMPARED TO INFECTION)
Abortion	RR=2 to 2.5 [1.7-3.8] *(Lopez Gatius, et al., 2012; Ordronneau, 2012)* *RR applied to seropositive cows and heifers with the abortion prevalence when no infection as reference (Lopez Gatius, 2014 ; Ordronneau, 2012)*	RR=0.69 [0.453-1.06] (p=0.09) *(Ordronneau, 2012)* *RR applied to seropositive cows and heifers with the abortion prevalence when infection as reference (Lopez Gatius et al., 2012; Ordronneau, 2012)*
Placental retention	RR=1.52 [1.06-2.19] *(Ordronneau, 2012)* *RR applied to seropositive cows only with the placental retention prevalence when no infection as reference*	No *(Ordronneau, 2012)*
Metritis	RR=2.5 *(Valla, et al., 2014)* *RR applied to seropositive cows only with the metritis prevalence when no infection as reference*	No *(Ordronneau, 2012)*
Late AI after given AI (27-90 d)	*RR= 1.85 for both heifers and cows (extrapolation from vaccine impact) Expressed as 2 d. extra CCI per seropositive cow and heifer*	RR=0.538 [0.30-0.96] for heifers RR=1 for cows *(Ordronneau, 2012)* Same impact for cows that remain seropositive (infected) after vaccination
Pregnancy and fertility (FSCR, extra service and CCI expected to be additive)	FSCR: -50% of change (RR for preg≈0.5) *(Lopez Gatius, et al., 2012) Expressed as 3 d. extra CCI per seropositive cow and heifer*	Same impact for cows and heifers that remain seropositive (infected) after vaccination
	+ 0.4 service per conception (extrapolation from benefits of vaccination ; Lopez Gatius, et al., 2012) Applied for seropositive cow and heifer	Same impact for cows and heifers that remain seropositive (infected) after vaccination
	CCI: +14 d (extrapolation from benefits of vaccination ; Lopez-Helguera et al, 2014). Applied for seropositive cows only	Same impact for cows that remain seropositive (infected) after vaccination

Table 1 Impact in cases of QF infections and in cases of vaccination

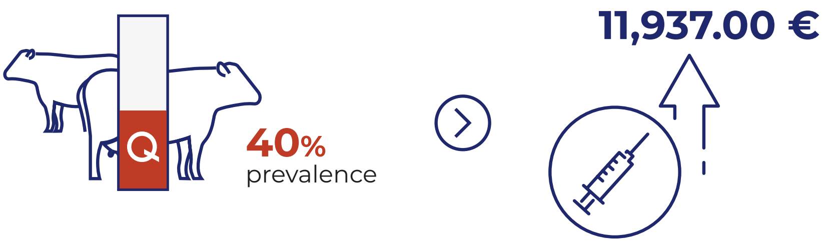
Blue: applications in the models
Orange: extrapolations because of limited data available

Base prevalence (out of Q fever) for abortion, placental retention and metritis (puerperal metritis and clinical endometritis) were set at **5%**, **4%** and **9 %**, respectively *(Santos, 2004, De Vries, 2006)*.

4. A practical case

Finally, the authors applied and tested the model *(Raboisson et al., 2022)* in a 100-cows dairy herd, considering two different levels of Q fever prevalence.

For a 100 cows herd with a low prevalence of Q fever infection (20%) before vaccination, the 3-year cumulative profit is estimated at 3,169.00 €.

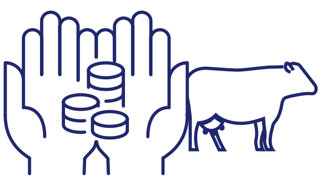


For a herd with a higher prevalence (40%) of seropositive cows before vaccination, the cumulative profit is 11,937.00 €.

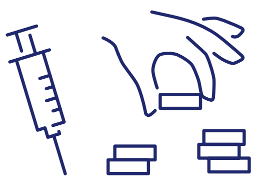
5. Conclusions

Q fever as a major impact in the reproductive performance dairy herds, resulting in clinical signs such as abortion, metritis, retained placenta and infertility.

Coxevac calculator has been developed to:



1. Estimate the costs of the disease on a given dairy herd



2. Calculate the economic benefit of vaccinating against Q fever and the return on investment of Coxevac.

In an effort to create a user-friendly tool, we willingly decided to focus the data only on the major consequences of the disease for cattle, i.e. reproductive disorders. Therefore, we can consider that the ROI is underestimated, and additional tangible and non-tangible benefits are expected.

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Notes